

ML Practical Test Report – Syndrome Classification

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1 Methodology

The first step consisted of loading the dataset from the provided pickle file and transforming it into a structured DataFrame format, which enables efficient manipulation and analysis.

After that, an exploratory data analysis (EDA) was conducted to examine the general characteristics of the dataset and ensure data integrity, including checks for missing values, consistency of embedding dimensions, and potential duplicates.

To better understand the distribution and structure of the embeddings, the t-SNE (t-distributed Stochastic Neighbor Embedding) technique was applied to reduce the high-dimensional vectors into a two-dimensional space, allowing for visual inspection of potential clustering patterns.

Next, K-Nearest Neighbors (KNN) models were implemented to determine the optimal number of neighbors (k) and the most suitable distance metric (Euclidean or Cosine). This was achieved through hyperparameter tuning using a range of k values from 1 to 15, evaluated using the F1-score as the primary metric.

Stratified cross-validation was used to preserve the class distribution across folds, ensuring a more reliable and robust evaluation of model performance.

After identifying the optimal configuration (k and distance metric), a final model was trained and evaluated using multiple metrics, including F1-score, AUC (Area Under the Curve), and Top-3 Accuracy. The use of Top-k accuracy is particularly relevant in medical applications, where models often serve as decision-support tools rather than definitive classifiers.

Finally, ROC curves were generated and averaged across cross-validation folds to visually compare the performance of the Euclidean and Cosine distance metrics.

2 Results

The results were divided into four main parts:

2.1 EDA and Integrity Checks

The first part is the data flattened into a tabular structure, where each row represents a single image embedding associated with its syndrome and subject, and the outputs from the EDA and data integrity checks, presented in Figures 1, 2, and 3:

```
===== DATASET OVERVIEW =====
(1116, 4)
  syndrome_id  ...                               embeddings
0   300000082  ...  [-0.03718013, 1.741486, 1.2061033, -0.45145318...
1   300000082  ...  [2.4249947, 0.17991967, 2.9503245, 0.37993023,...
2   300000082  ...  [-1.7893314, -0.21621145, 0.43431538, -0.55232...
3   300000082  ...  [-1.1436926, -0.7146209, 0.31948757, 0.4556819...
4   300000082  ...  [0.81417066, 1.3274913, 0.84728503, -0.2123311...

[5 rows x 4 columns]
```

Figure 1 – Dataset overview

```
===== BASIC STATS =====
Syndromes: 10
Subjects: 941

===== IMAGES PER SUBJECT =====
count    941.000000
mean      1.185972
std       0.829288
min       1.000000
25%       1.000000
50%       1.000000
75%       1.000000
max      11.000000
dtype: float64
```

Figure 2 – Basic stats and images per subject

```
===== DATA INTEGRITY =====  
Missing:  
  syndrome_id    0  
  subject_id     0  
  image_id       0  
  embeddings      0  
dtype: int64  
Embedding sizes:  
  embeddings  
320    1116  
Name: count, dtype: int64  
Duplicates:  
  0  
  
===== SUBJECT-SYNDROME RELATION =====  
syndrome_id  
1    941  
Name: count, dtype: int64
```

Figure 3 – Data integrity and subject-syndrome relation

2.2 Data Visualization using t-SNE

The second part consists of the visualization of the embeddings using t-SNE, shown in Figure 4:

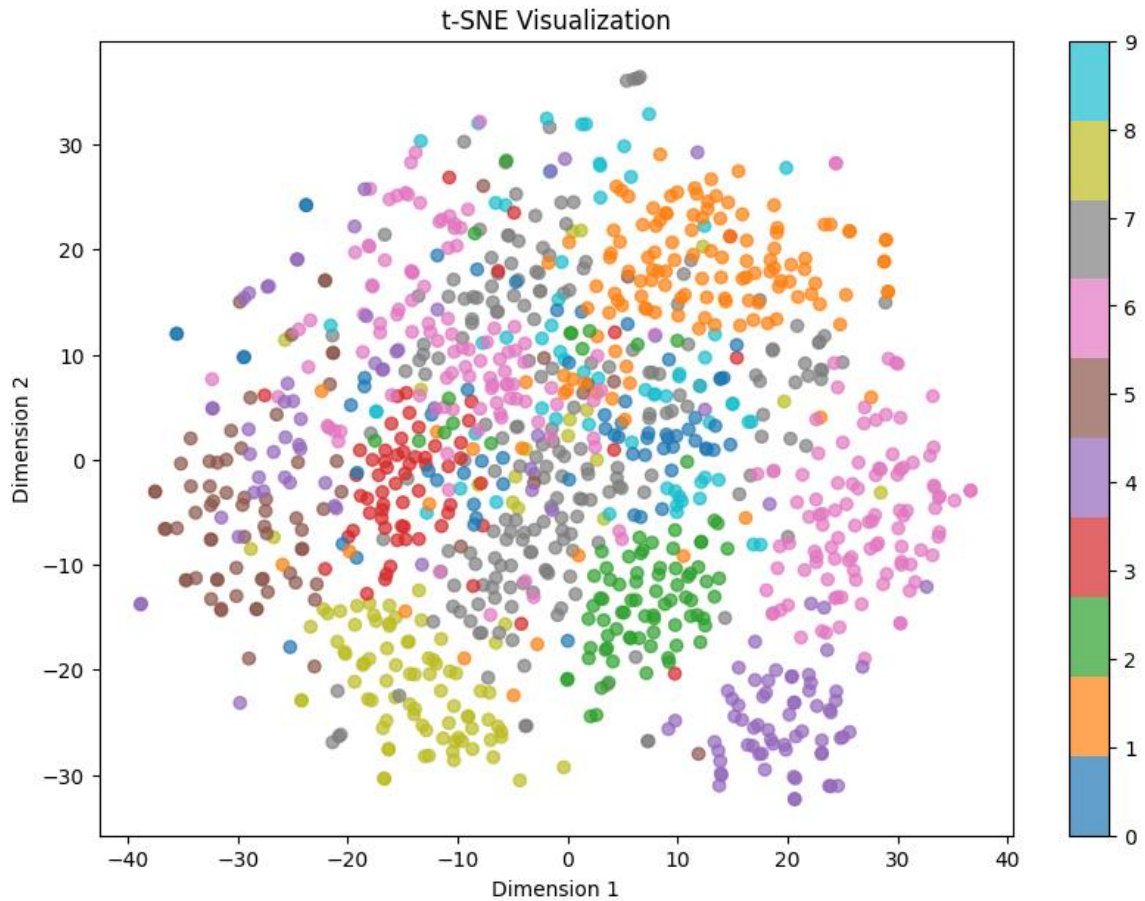


Figure 4 – t-SNE visualization

2.3 Optimal k Value

The third part presents the analysis of F1-scores across different values of k and distance metrics (Figure 5). The first column corresponds to the k values, the second to the F1-scores using the Euclidean distance, the third to the F1-scores using the Cosine distance, and the fourth to the best F1-score for each k . The table is sorted in descending order based on the best F1-score.

k	euclidean_f1	cosine_f1	best_f1
7.0	0.713	0.801	0.801
8.0	0.73	0.793	0.793
9.0	0.73	0.79	0.79
13.0	0.745	0.788	0.788
15.0	0.755	0.787	0.787
6.0	0.695	0.786	0.786
12.0	0.732	0.784	0.784
11.0	0.736	0.783	0.783
10.0	0.726	0.783	0.783
14.0	0.751	0.781	0.781
5.0	0.689	0.779	0.779
4.0	0.683	0.758	0.758
3.0	0.66	0.745	0.745
1.0	0.655	0.704	0.704
2.0	0.616	0.682	0.682

Figure 5 – Optimal K analysis

2.4 Model Evaluation Results and ROC Curves Comparison

The fourth part includes the model evaluation results. Figure 6 summarizes the final performance metrics of the best-performing model. The first column contains the metric names, while the second and third columns present the results for the Euclidean and Cosine distance metrics, respectively.

metric	euclidean	cosine
f1	0.713	0.801
auc	0.927	0.953
top3	0.894	0.929

Figure 6 – Final model metrics

Finally, Figure 7 shows the ROC curves for both distance metrics, along with a baseline random classifier (diagonal line) representing a model with no discriminative power.

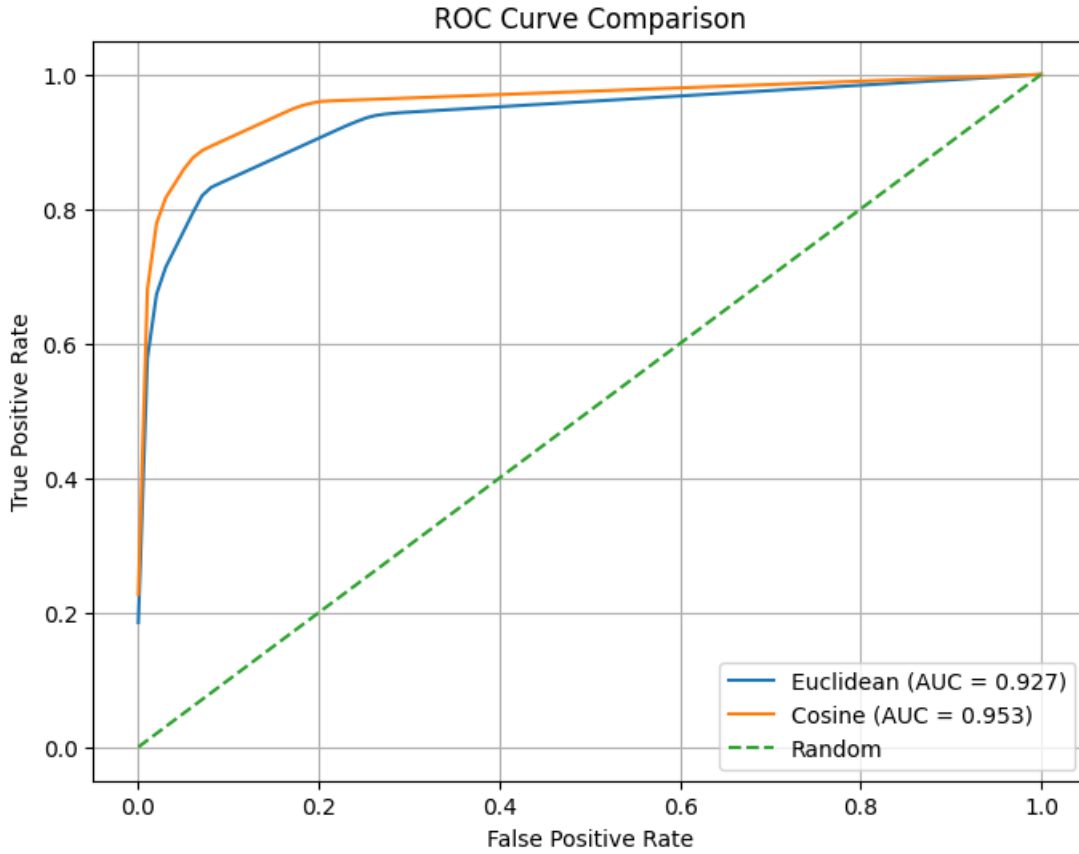


Figure 7 – ROC Curves comparison

These results provide the basis for the comparative analysis between distance metrics and model performance.

3 Analysis

3.1 EDA and Integrity Checks Analysis

From Figure 1, the dataset contains 1116 samples, corresponding to 1116 image embeddings.

Figure 2 shows that the dataset includes 10 distinct syndromes and 941 subjects. Each sample is represented by a 320-dimensional embedding vector.

The analysis of images per subject indicates that most subjects have only one image (mean ≈ 1.19), with a small number of outliers reaching up to 11 images. This distribution suggests that the risk of data leakage due to multiple images per subject is limited, but still relevant to consider.

From Figure 3, the dataset is fully consistent: no missing values were found, all embeddings have a fixed dimensionality of 320, and no duplicated entries were identified. Additionally, each subject is uniquely associated with a single syndrome, confirming a one-to-one relationship between `subject_id` and `syndrome_id`.

Overall, the dataset is clean, well-structured, and suitable for modeling.

3.2 Data Visualization Analysis

From Figure 4, the t-SNE visualization reveals the presence of well-defined clusters for several syndromes, indicating that the embeddings capture meaningful class-specific patterns. However, overlapping regions are also observed, suggesting that some syndromes share similar feature representations. This indicates that the classification task is not trivial and that some level of misclassification is expected.

Clusters located at the periphery of the embedding space tend to be more separable, while central regions exhibit higher class mixing, indicating ambiguity between certain classes. It is also possible to observe class imbalance in the visualization, with some syndromes having significantly more samples than others. This further supports the use of stratified cross-validation.

Although t-SNE is useful for visualization, it primarily preserves local structure and may distort global relationships (Géron). Therefore, the visualization should be interpreted qualitatively rather than as an exact representation of class separability.

Overall, the embeddings exhibit meaningful structure, but with overlap that inherently limits classification performance. This makes KNN a reasonable baseline model, while also indicating that more complex models could further improve results.

3.3 Optimal k Value Analysis

From Figure 5, cosine distance consistently outperforms Euclidean distance across all tested values of k . The best performance is achieved with cosine distance at $k = 7$, reaching an F1-score of approximately 0.801. In comparison, Euclidean distance achieves its best result at $k = 15$ with an F1-score of approximately 0.755. This suggests that the embedding space encodes information primarily through vector orientation rather than magnitude, making cosine similarity more appropriate for this task.

The observed performance is consistent with the t-SNE visualization: although clusters are partially separable, the presence of overlapping regions leads to an inherent classification error of around 20%, particularly in regions where classes are less distinct.

Overall, the results indicate that the embeddings contain meaningful structure, but are not perfectly separable.

3.4 Model Evaluation Analysis

The Table presented in Figure 6 shows consistent superiority of cosine distance across all metrics:

The F1-score reaches approximately 0.801 with cosine distance, compared to approximately 0.713 with Euclidean distance. This indicates Strong classification performance, with remaining errors mainly driven by class overlap.

The AUC values are particularly high (0.953 for cosine and 0.927 for Euclidean), indicating strong class separability in the high-dimensional space. This suggests that, despite overlap in lower-dimensional projections, the model is able to effectively distinguish between classes.

Top-3 accuracy further reinforces this result, reaching approximately 92.9% with cosine distance. This means that even when the model makes an incorrect top prediction, the correct class is often among the top candidates. This is particularly relevant in medical applications, where models are typically used as decision-support tools rather than definitive classifiers.

3.5 ROC Curves Comparison Analysis

From Figure 7, both ROC curves are significantly above the random baseline, confirming strong predictive performance. The cosine curve consistently dominates the Euclidean curve across all thresholds, reinforcing its superiority as a distance metric for this task. Both curves are close to the top-left corner, indicating high true positive rates combined with low false positive rates. This reflects the model's ability to effectively distinguish between classes.

An important observation is that, despite the overlap observed in the t-SNE visualization, the model achieves high AUC values. This indicates that separability in the original high-dimensional space is stronger than what is suggested by the 2D projection.

3.6 Results Analysis Summary

Overall, cosine distance significantly outperforms Euclidean distance across all evaluation metrics, confirming its suitability for high-dimensional embedding spaces.

The model achieves an F1-score of approximately 0.80 and an AUC above 0.95, indicating strong classification performance. Additionally, the high Top-3 accuracy (~93%) shows that even incorrect predictions frequently include the correct class among the top candidates.

4 Challenges and Solutions

One challenge identified during the analysis was the imbalance in the number of samples across syndromes. Some classes contain significantly more embeddings than others, which could bias model evaluation. To address this, stratified cross-validation was used to preserve class distribution across folds, ensuring a more reliable evaluation. Additionally, the presence of multiple images per subject introduced a potential risk of intra-subject correlation. Although most subjects have only one image, this issue was also mitigated by using stratified cross-validation, ensuring that model evaluation remains robust.

Another initial concern was whether `subject_id` values were local to each syndrome or globally unique. If subject identifiers were reused across syndromes, this could lead to incorrect assumptions and potential data leakage. However, exploratory analysis confirmed a one-to-one relationship between subjects and syndromes, ensuring consistency and eliminating this risk.

During visualization, an issue arose with label representation in the t-SNE plot. Using continuous values for syndrome labels resulted in a misleading color gradient, making class separation difficult to interpret. This was resolved by encoding the labels into discrete categories, allowing for clear and interpretable cluster visualization. Another challenge relates to the stochastic nature of t-SNE. Different runs may produce visually different projections due to variations in initialization and library versions. However, this does not affect the validity of the analysis, as t-SNE preserves local neighborhood structure. Therefore, interpretation was based on relative clustering patterns rather than absolute positioning.

Finally, selecting an appropriate distance metric for high-dimensional embeddings was a key challenge. Euclidean distance may not capture meaningful relationships in such spaces due to the curse of dimensionality. This was addressed by evaluating both Euclidean and cosine distances, with cosine distance demonstrating superior performance.

5 Recommendations

First, more advanced models could be applied to better capture complex decision boundaries. Since the embeddings exhibit partial overlap between classes, models such as Support Vector Machines (SVM) with non-linear kernels or neural networks could improve classification performance by learning more flexible boundaries than KNN.

Additionally, although stratified cross-validation was used, a small degree of data leakage may still occur due to multiple images from the same subject appearing in both training and test sets. A more robust validation strategy would involve grouping samples

by subject_id (using GroupKFold), ensuring that all images from a given subject are kept within the same fold.

Finally, increasing the dataset size and improving class balance could help reduce overlap between syndromes, leading to more accurate and robust models.